

Dr. Zoe Molitor (she or they)

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Appointments

Boston College — Earth and Environmental Sciences , Postdoctoral Researcher	Chestnut Hill, MA Jan 2026 – present
Yale University — Earth and Planetary Sciences (EPS) , NSF EAR Postdoctoral Fellow	New Haven, CT Jan 2024 – Jan 2026

Education

PhD MIT — Earth, Atmospheric and Planetary Sciences (EAPS) , Geology	Cambridge, MA Jan 2018 – Jan 2024
BS Purdue University , Geology and Geophysics	West Lafayette, IN Jan 2015 – Jan 2018

Publications

Extreme Moho Relief Preserved in Ohio's Crust Z. Molitor , F. Link, M. Long, J. Ague (GSA Today, 36, 4–9)	Jan 2026
A whole-rock geochemistry-based semi-quantitative proxy for magmatic oxygen fugacity Z. Liu, O. Jagoutz, H. Rezeau, H. L. L. Zhang, B. Klein, Z. Molitor , et al., D. C. Zhu (Earth and Planetary Science Letters, 673, 119742)	Jan 2026
Bedrock Geology of NE Connecticut: A Transect from the Bronson Hill Anticlinorium to Putnam-Nashoba Terrane Z. Molitor , J. Ague, B. Graham digitalmaine.com/geo_docs/204 (In A.D. Rooney & M.D. Long (Eds.), Guidebook for Field Trips in Southwestern Connecticut (Trip D, pp. 86–112), Geology Documents)	Jan 2025
Permo-Carboniferous crustal structure and state of strain in the New England Appalachians, USA Z. Molitor , O. Jagoutz, A. Moser, A. Cruz-Urbe, A. Cross (Geosphere, 21(6), 991–1019)	Jan 2025
Crustal section in the New England Appalachians records polybaric fractional crystallization-driven magma differentiation Z. Liu, O. Jagoutz, Z. Molitor , J. Ramezani, D. C. Zhu (Earth and Planetary Science Letters, 671, 119647)	Jan 2025
The viscosity of a partially molten layer in a paleo-orogenic plateau Z. Molitor , T. Mittal, O. Jagoutz (Earth and Planetary Science Letters, 648, 119060)	Jan 2024
ReversePetrogen: A Multiphase Dry Reverse Fractional Crystallization-Mantle Melting Thermobarometer Applied to 13,589 Mid-Ocean Ridge Basalt Glasses S. Brown Krein, Z. Molitor , T. L. Grove (Journal of Geophysical Research: Solid Earth, 126(8), e2020JB021292)	Jan 2021

Testing models of orogenic conjugate shear in the New England Appalachians

Z. Molitor, O. Jagoutz, A. Cross, A. Moser
(In preparation)

Invited Talks

Extreme Moho Relief Beneath the State of Ohio

Lahmont-Doherty Earth Observatory,
Geodynamics Seminar
Nov 2025

Conference Abstracts

Spatio-Temporal Controls on Orogenic Thermal Structure and Evolution in the Southern New England Appalachians

22–23 Mar 2026

Z. Molitor, J. Ague, B. Graham, J. Eckert, A. Cruz-Urbe
(Abstract 29-5, 2026 Northeast GSA)

Assessing the Impact of High Temperature Metamorphism and Partial Melting on the Viscosity of the Tattnall Hill Formation

19–22 Oct 2025

B. Graham, **Z. Molitor**, J. Ague
(Abstract 284-7, 2025 GSA Connects)

Spatio-Temporal Controls on Orogenic Thermal Structure and Evolution in the Southern New England Appalachians

19–22 Oct 2025

Z. Molitor, J. Ague, B. Graham, J. Eckert
(Abstract 284-7, 2025 GSA Connects)

Constraining the 3D Geometry of Conjugate Orogenic Strike-Slip Shear Zones in the New England Appalachians

17–19 Mar 2024

Z. Molitor, O. Jagoutz, A. Cross, A. Cruz-Urbe, A. Moser
(Abstract 20-6, 2024 Northeast GSA)

Testing Models of Conjugate Faulting in the Late Paleozoic New England Appalachian Orogen

12–16 Dec 2022

Z. Molitor, L. Royden, O. Jagoutz
(Abstract T55B-0068, 2022 AGU Fall Meeting)

A New Model for Dynamic and Thermal Compensation around Mantle Plumes: Implications for Volume and Heat Flux

13–17 Dec 2021

Z. Molitor, L. Royden, O. Jagoutz
(Abstract DI25B-0027, 2021 AGU Fall Meeting)

Mantle Plume – Spreading Ridge Interaction: A Comparative Study of the Red Sea and Reykjanes Ridges

4–8 May 2020

Z. Molitor, O. Jagoutz, L. Royden, S. Brown, G. Port, A. Dogru, J. Keller
(Abstract EGU2020-11138, 2020 EGU General Assembly)

Teaching Experience and Outreach

Adjunct Professor — Structural Geology, EESC 3385

Boston College
Fall 2026

Graduate Teaching Assistant, 12.002

MIT
Spring 2021 & Spring 2022

Undergraduate Teaching Assistant, EAPS 12000

Purdue University
Jan 2017

Lead Organizer, Earth Science Week-Long Workshop — Pathways Summer Scholars	Jan 2026
Co-Organizer, Earth Science Week-Long Workshop — Pathways Summer Scholars High School Program	Yale University Jan 2025
Group Leader, Chemistry Rotation During West Campus Science Festival	Yale University Jan 2024
Administrative Member, LINK12 K-12 Outreach Program	EAPS, MIT Jan 2022 – Jan 2024
Letters to a PreScientist — Pen Pal Program with K-12 Students	Jan 2022
Field Trip on the Geology of the White Mountains, DEAPS Freshman Orientation	Jan 2022
Boston826: Hydraulic Fracturing Demonstration and Interview with Local K-12 Students	Jan 2021
Field Trip on the Geology of Yellowstone, DEAPS Freshman Orientation	Jan 2020

Service

Founder and Lead Organizer, LGBTQ+ Working Group	Yale University Jan 2024 – Jan 2026
WEPSY (Women and Nonbinary People in EPS) Organizing Committee	Yale University Jan 2024 – Jan 2026
Member, EPS Department IDEA Committee	Yale University Jan 2024 – Jan 2026

Mentoring

Brynn Graham (Undergraduate, Mt. Holyoke College), SURF Program	Yale University Summer 2025 – Spring 2026
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Field Experience

Spring 2026: Nahant and Salem, MA — 3 Days
Fall 2025: NEIGC, Connecticut — Field Trip Leader
Fall 2024: Connecticut, Massachusetts, and New Hampshire — 6 Weeks
Jun 2023: Sicily, Italy (MIT Department Field Trip) — 1 Week
Summer 2022: New England Appalachians (NH, MA, CT) — 4 Weeks
Jan 2022: Mojave Desert (MIT Field Camp) — 2 Weeks
Dec 2021: Vermont and New Hampshire — 1 Week
Nov 2021: Newfoundland, Canada — 2 Weeks
Jun 2021: New Hampshire — 1 Week
Fall 2020: New Hampshire — 4 Weeks
Jan 2019: Mojave Desert (MIT Field Camp) — 2 Weeks
Jun 2018: Sardinia, Italy (Undergraduate Field Camp) — 4 Weeks
Mar 2018: Utah (Undergraduate Field Methods) — 1 Week

Technical Skills

Programming Languages: Python, MATLAB (MTEX)

Computer Programs: ArcGIS, Adobe Illustrator, Inkscape, Perple_X, TheriakDomino, Microsoft Office Suite

Laboratory Analysis: SEM/EPMA (Electron Probe Microanalyzer), SEM/EBSD (Electron Backscatter Diffraction), Heavy Mineral Separation, LA-ICP-MS